

The Utility Problem

Traditionally, electricity in the grid has flowed from the “top” of circuits (the high voltage substations connecting the distribution grid with the transmission grid) “down” to “consumers” such as residences, businesses or industrial facilities. With the addition of DERs such as solar (PV), batteries and electric vehicles (V2G), electricity flow can become bi-directional (temporarily reversed, aka upstream power flow). This reversed power flow happens when the sum of distributed generation is higher than the sum of consumption, all given in some specific supply area of the grid. Such an occurrence is increasingly common as ever larger quantities of solar microgeneration (PV) are connected to the grid. For many utilities, the reversed power flow, especially on the edge of the low voltage (LV) and medium voltage (MV) grid, is not a desirable situation. This situation puts increased constraints on MV grid voltage regulation and influences the size of fault currents in the MV, thus making grid protection device settings more challenging.

This use case is aimed to simulate and quantify the additional amount of PV generation (PV capacity) allowed on each LV grid before reaching a state of reversed power flow (reversed power flow = power flow in the direction of LV grid to MV grid). For the purpose of this analysis, the LV grid is defined as a supply area which is served by one MV/LV transformer.

The result of such analysis might indicate such situations and can give valuable insights for decision-making on new PV generation connection permits and/or can be a good source of data for grid upgrades or grid planning activities.

The Data Solutions and Methods

For conducting such an analysis, it is necessary to have several datasets ready. These datasets are as follows:

1. Grid areas for analysis - these are LV grids which are defined as supply areas supplied by MV/LV transformer. This data should contain all consumers and prosumers in the areas.
2. Most common solar PV generation configurations - this will be used for simulation purposes.
3. Consumption and generation or net load time series from areas that will be analyzed.
4. Standard PV generation time-series - for simulation purposes.

In addition, the above-mentioned datasets should be synchronized and aligned in time and space so that they create a unified data layer. The Awesense Energy Transition Platform offers these capabilities. Awesense Platform’s unique linkage of geospatial data (GIS) with time-series data enables this type of analysis. With data populated in the Awesense Platform, the Energy Data Model’s (EDM) database views and functions will accelerate the building of the Reversed Power Flow Analysis for PV Generation.

To get the necessary datasets for the Reversed Power Flow Analysis for PV Generation, the Awesense EDM offers multiple methods:

1. Grid areas for analysis:

- a. This comes from the core geospatial dataset for Awesense Energy Transition Platform - the GIS data. GIS data, together with its connectivity information, provide all necessary information for the definition of areas for analysis. Awesense Energy Data Model's (EDM) unique geospatial connectivity function can be used to gather all necessary data simply.
2. Most common solar PV generation configurations:
 - a. This information can be obtained from Awesense EDM by analyzing the attributes of the most common photovoltaic GIS elements. Photovoltaics are grid elements which are fully modelled within the Awesense EDM.
 - b. The second option is to choose your own photovoltaic configuration for these analyses.
3. Consumption and generation (or net load) time series:
 - a. Awesense EDM models the time-series geospatially, which means that all time-series which are integrated within the platform can be easily queried. This is no different for meter load time series or separate consumption and generation time series. After the areas for analysis are defined, the geospatial element IDs from these areas can be used for querying the time series data.
4. Standard PV generation time series:
 - a. These can be obtained in two ways: either via mathematical formulas describing solar generation at a particular location and time under ideal conditions or by determining "typical" amounts based on actual time series associated with existing PV elements. For the latter approach, as in the step before, the Awesense EDM offers easy querying of time series. To get this data, it is necessary to filter for PV elements, retrieve their consumption time series and then aggregate them in some statistical manner to produce typical generation time series curves. Regardless of the approach, the standard curves can then be used to simulate additional PV generation elements in the analyzed part of the grid.

To put this all together, the analyst should be able to quantify the net power flow for each period of time in the analyzed area. To do this, it is necessary to sum the net load over consumers and producers in the entire area; these net loads may either be directly available from meters, or it may be necessary to sum all consumption and subtract all generation over time. If capacity exists, each area can be tested for different types and multiplications of the most common PV generation time series curves. Such analysis can be conducted on a per area basis, or such analysis can be created as a grid-wide analysis.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.

Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub [link](#) to the Jupyter notebook can be provided upon request.